Categorial granularity in the L1 development of the CP

A new argument from bilingualism

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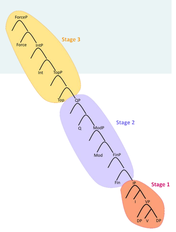
A new approach to the acquisition of the CP

Cartography (Rizzi, 1997; Cinque, 1999)**:** CP as a cover-term for an internally articulated functional domain, with multiple distinct positions; e.g., CP ≈ ForceP > TopP* > FocP > TopP* > FinP

Important language variation: not all heads overtly realized, and potentially syncretized in some languages (Giorgi & Pianesi, 1997; Bobaljik & Thráinsson, 1997; Cormany, 2015; Hsu, 2017).

→ How do children acquire left-peripheral structure?

Q1 Timing: How early is the CP acquired?
Q2. Directionality: Are CP projections acquired in a fixed order?



- **Maturation:** the CP matures universally late, after TP and VP. *Growing Trees Hypothesis* (Friedmann et al., 2021): two-stage development
- **Continuity:** CP universally and continuously accessible (e.g., Boser et al., 1992; Hyams, 1992; Poeppel & Wexler, 1993; Westergaard, 2009; i.a.).

Q3. Granularity: How structurally elaborate is the child's CP at each stage?

Fixed granularity (existing work)
Functional categories, and their degree of specification, are fixed by a universal functional sequence at all stages.

Flexible granularity (our proposal)
Initial access to a basic undifferentiated CP, which is 'split' at a later stage if input requires it.

Our questions: is evidence for flexible granularity detectable in production data? If so, does this vary depending on the typology of the language(s) children acquire?

Methodology

N = 11 — **Romance-based pairs**
6 monolinguals: Catalan, Spanish, Italian
5 Catalan-Spanish bilinguals

N = 8 — **Germanic-Romance pairs**
German-Italian, German-Spanish, Dutch-Italian, English-Spanish

CP diagnostics (Romance & Germanic)
→ Is there evidence for the CP at all?

- Wh-questions
- Yes/no-questions (Germanic only)
- V-to-C movement (Germanic only)
- Topicalisations
- Focalisations
- Illocutionary (main clause) complementisers (Romance only)
- Finite embedding

Split CP diagnostics (Romance)
→ Are multiple positions used?

- Top > Wh
- Top > Top/Foc (e.g., recursive topics)
- Complementiser > Wh/Top
- *Sí que/sí che* 'yes that' and *que sí que* 'that yes that' structures.
- Other possible combinations

E.g., Spanish ***El coche, dónde*** *lo compraste?* ('The car, where did you buy (it)?)

Study 1: A recurring asymmetry

Empirical precedent (Bosch & Biberauer, 2024; Bosch, 2025)**:** no evidence of access to more than one left peripheral projection until later developmental stages.

- **Early CP structures (including structurally high elements).**

→ Late Split CP-structures: productions requiring multiple left-peripheral projections emerge later, often abruptly.

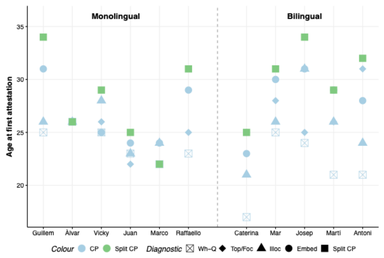


Fig 1. Age of emergence by diagnostic.

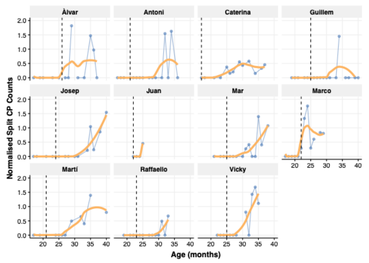


Fig 2. Split CP production by child.

Study 2: Selective underproduction

Nuancing the generalisation: distinct profiles across bilingual groups.

Germanic-Romance bilinguals show **superficially similar** trajectories.

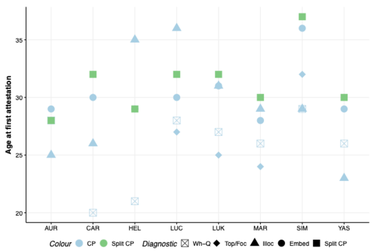


Fig 3. Age of emergence by diagnostic.

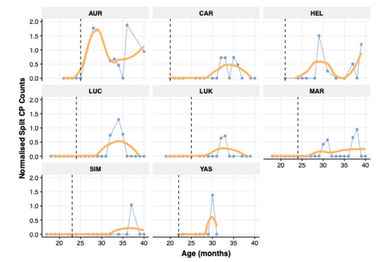


Fig 5. Split CP production by child.

However, Germanic-Romance bilinguals ***selectively underproduce*** Split CP-diagnostics!

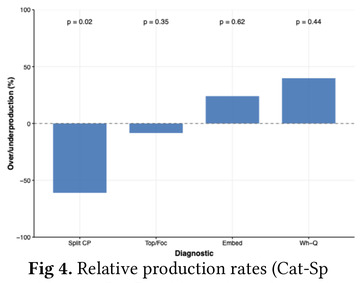


Fig 4. Relative production rates (Cat-Sp biling vs. Ger-Ro bilingual) by diagnostic.

Individual trajectories also qualitatively different → Fig 2 vs. Fig 5

- ☐ Romance-only learners: **steadily increasing production** of Split CP structures (exponential or sigmoidal)
- ☐ Ger-Ro bilinguals: **stagnation**, intermittent peaks and declines.
  - ☐ *Exception*: AUR, the only unbalanced bilingual.

Granularity as a developmental variable

Implications

- **Fixed granularity approaches under- or overgenerate.**
  - Early CP-structures only compatible with continuity approaches.
  - Neither bottom-up maturation nor continuity can generate granularization.
- **Alternative explanations do not (fully) account for the data:**

✗ Input frequency — ✗ Sentence length — ✗ Derivational complexity

Ask us for more details!

Analysis

Our proposal: children's categorial space is ***flexible***, and some functional distinctions are learned.

1. An emergent cartography

Crosslinguistic variation: articulated 'split' CP in languages like Italian, Catalan or Spanish (Rizzi, 1997); single-head, syncretised CP in English, German or Dutch (Hsu, 2017; Walkden, 2017).

Granularization: basic single-head CP → split CP.

2. A Maximize Minimal Means approach (Biberauer, 2019)

Feature Economy: minimize postulation of new formal features.

Feature Generalization: maximize already-acquired features.

→ Pressure to maximize representations shared across L1s – a **structure sharing bias**

*(1) + (2) predict the **direction** of crosslinguistic influence (CLI)*

Granularity-mismatch hypothesis on CLI
When their two languages differ in CP-granularity, bilinguals favour the retention of shared, developmentally prior structures (a basic CP) over further granularization.

In a nutshell

TARGET GRAMMARS

English, German, Dutch
Single syncretized C head
CP

Catalan, Spanish, Italian
Cartographic-type split CP
ForceP > TopP* > FocP > FinP

PROPOSED ACQUISITION PROCESS

Granularity match
Romance monolinguals, Cat-Sp bilinguals
basic CP → split CP

Granularity mismatch
Germanic-Romance bilinguals
basic CP --DELAYED--> split CP

1. **CP starts undifferentiated** and splits later
2. Splitting is **input-driven**: granularity is emergent and crosslinguistically variable
3. **Mismatched L1s delay splitting**

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